

Polarized Membrane Cosmology, Opus I: Branch-Relative Matter Asymmetry and a Topological Relic Channel

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Abstract

This first note in a proposed series formulates a polarized-membrane toy model in which an observed matter–antimatter asymmetry is branch-relative rather than global. The basic projection is not a binary sorting map, but a three-sector cosmological yield

$$\mathcal{H}_{\text{deep}} \longrightarrow \mathcal{H}_A \oplus \mathcal{H}_B \oplus \mathcal{H}_{\mathcal{R}},$$

where branch A carries a matter-oriented visible sector, branch B carries the conjugate orientation, and $\mathcal{R}_{\Pi}$ denotes residual topological excitations of the branch-separating order parameter. The model does not claim to generate a nonzero global baryon number. Instead, it asks whether a globally balanced structure can yield a matter-dominated observed branch while depositing part of the projection stress-energy into neutral, stable, hidden-sector relics. This reframes the dark matter abundance, $\Omega_{\text{DM}}/\Omega_b \simeq 5.3$, as a target for the same projection event that fixes the branch baryon residue. The construction is deliberately presented as incomplete. It becomes physical only if the membrane operator Π , its projection scale T_{Π} , and the relic yield $\rho_{\mathcal{R}}/\rho_b$ are derived from geometry, topology, or dynamics. Local Standard Model gauge physics is preserved: the branch projection is not a charge-superselection rule, positrons and pair annihilation remain allowed, and all long-range Standard Model gauge charges must be screened or neutralized inside each branch.

1 Purpose of Opus I

The observed universe contains a small baryon residue relative to the photon bath,

$$\eta_B = \frac{n_B - n_{\bar{B}}}{n_{\gamma}} \sim 6 \times 10^{-10}, \quad (1)$$

and roughly five times more dark matter than baryonic matter,

$$\frac{\Omega_{\text{DM}}}{\Omega_b} \simeq \frac{\Omega_c}{\Omega_b} \approx 5.3. \quad (2)$$

In conventional cosmology these two facts are usually treated as outputs of different mechanisms: baryogenesis or leptogenesis for the baryon asymmetry, and freeze-out, misalignment, hidden-sector dynamics, or another mechanism for dark matter.

This note explores a different bookkeeping architecture. It does not attempt to replace established baryogenesis with a completed dynamical theory. Instead, it asks whether a single membrane transition can define three correlated outputs:

$$\text{visible branch } A, \quad \text{conjugate branch } B, \quad \text{neutral relic sector } \mathcal{R}_{\text{II}}. \quad (3)$$

The ambition is not merely to hide antimatter. The useful version must instead relate two observed branch quantities,

$$\eta_A \sim 6 \times 10^{-10}, \quad \frac{\rho_{\mathcal{R}}}{\rho_{b,A}} \approx 5.3, \quad (4)$$

to one projection operator and one projection scale. If those quantities are fitted independently, the model has no predictive advantage.

The central claim of Opus I is therefore:

A polarized membrane projection may encode branch-relative matter asymmetry and a dark relic channel, but it becomes physics only if the projector, projection scale, and relic yield are derived rather than postulated.

2 From binary sorting to three-sector yield

A purely binary membrane picture would write

$$\mathcal{H}_{\text{deep}} \longrightarrow \mathcal{H}_A \oplus \mathcal{H}_B, \quad (5)$$

where A receives matter-oriented histories and B receives conjugate histories. That version is too weak. It merely relocates the compensating sector unless it predicts a number or a signal.

The upgraded architecture allows the membrane transition to leave a residual component:

$$\mathcal{H}_{\text{deep}} \longrightarrow \mathcal{H}_A \oplus \mathcal{H}_B \oplus \mathcal{H}_{\mathcal{R}}. \quad (6)$$

Here $\mathcal{H}_{\mathcal{R}}$ is not ordinary antimatter hidden behind a wall. It is a relic sector associated with the branch-separating order parameter. Schematically, the transition at T_{II} produces

- A : visible matter-oriented branch,
- B : conjugate branch,
- $\mathcal{R}_{\text{II}}$: neutral topological relics or compact hidden-sector defects.

The stress-energy budget after the projection is correspondingly

$$\rho_{\text{post}} = \rho_A + \rho_B + \rho_{\mathcal{R}}. \quad (7)$$

For observers in branch A , the potentially observable sector is

$$\rho_{\text{obs},A} = \rho_{b,A} + \rho_{\gamma,A} + \rho_{\mathcal{R},A} + \cdots, \quad (8)$$

where $\rho_{\mathcal{R},A}$ denotes the portion of the relic stress-energy gravitationally accessible to branch A .

This changes the model's target. The question is no longer only “why does branch A contain matter?” It becomes:

$$\text{Can one projection event explain both } \eta_A \text{ and } \Omega_{\text{DM}}/\Omega_b? \quad (9)$$

3 Membrane spectral structure

Let $\mathcal{H}_{\text{deep}}$ be a hypothetical higher-dimensional state space. The observed particle labels are treated as spectral labels of a membrane operator family rather than as primitive labels. A schematic decomposition is

$$\mathcal{H}_{\text{deep}} \longrightarrow \mathcal{H}_{\text{gen}} \otimes \mathcal{H}_{\text{orient}} \otimes \mathcal{H}_{\text{branch}} \oplus \mathcal{H}_{\mathcal{R}}. \quad (10)$$

The generation factor is meant only as a structural analogy. The electron, muon, and tau are not the same Standard Model particle; however, they may be viewed in a deeper theory as distinct resonant modes or eigenstates of an internal operator. Write

$$G|\ell_i\rangle = g_i|\ell_i\rangle, \quad i = e, \mu, \tau. \quad (11)$$

A string-like or Kaluza–Klein-like completion would attempt to derive the eigenvalues and masses rather than insert them.

The same membrane family is hypothesized to contain a branch-polarization operator Π . In the simplest two-branch limit,

$$P_A = \frac{1 + \Pi}{2}, \quad P_B = \frac{1 - \Pi}{2}. \quad (12)$$

With a relic channel, this must be generalized to three positive operators or projectors,

$$P_A + P_B + P_{\mathcal{R}} = I, \quad P_i P_j \simeq \delta_{ij} P_i \quad (13)$$

inside the post-transition effective description. The approximate sign is intentional: in an effective theory, branch leakage or portals may be small but nonzero.

Use basis states

$$|i, \sigma, X\rangle, \quad (14)$$

where i labels a generation or resonance mode, $\sigma = \pm$ labels orientation, and $X = A, B$ labels the branch. A strong projection limit is

$$P_A|i, +\rangle \simeq |i, +\rangle, \quad P_A|i, -\rangle \simeq 0, \quad (15)$$

$$P_B|i, -\rangle \simeq |i, -\rangle, \quad P_B|i, +\rangle \simeq 0. \quad (16)$$

These equations describe large-scale cosmological occupation measures, not local particle-production rules.

4 Local gauge neutrality is mandatory

The membrane cannot separate uncompensated Standard Model gauge charge. In particular, the branch must satisfy

$$Q_A \simeq 0, \quad Q_B \simeq 0, \quad Q_{\mathcal{R}}^{\text{SM}} = 0, \quad (17)$$

where Q denotes ordinary electromagnetic charge, with analogous color-neutrality constraints for QCD. This is not optional. Long-range unscreened Standard Model gauge fields would be immediately excluded by cosmology and by Gauss-law constraints.

Therefore the branch asymmetry is a baryon/lepton-orientation or history-occupation asymmetry, not a net electric charge separation. Branch A may have

$$B_A > 0, \quad Q_A = 0, \quad (18)$$

just as ordinary matter is electrically neutral on large scales because protons and electrons coexist. Branch B carries the conjugate orientation,

$$B_B = -B_A, \quad Q_B = 0, \quad (19)$$

if the total paired structure is globally balanced.

Local QED must remain intact in each branch:

$$\gamma\gamma \rightarrow e^+e^-, \quad (20)$$

$$e^+e^- \rightarrow \gamma\gamma. \quad (21)$$

Thus P_A is not a rule forbidding positrons in our branch. It is a statement about the cosmological background occupation of long-lived histories.

5 The relic sector as dark matter candidate

The relic channel $\mathcal{R}_\Pi$ is the main new ingredient in Opus I. The membrane projection may not map all pre-projection stress-energy cleanly into branches A and B . A residual component may remain as compact excitations of the branch-separating order parameter:

$$\mathcal{R}_\Pi = \text{localized knots, loops, solitons, or hidden flux tubes of } \Pi. \quad (22)$$

For this relic component to be a dark matter candidate, it must satisfy at least

$$Q_{\text{SM}}(\mathcal{R}_\Pi) = 0, \quad \text{Standard Model gauge neutrality,} \quad (23)$$

$$\tau_{\mathcal{R}} \gg t_0, \quad \text{cosmological stability,} \quad (24)$$

$$\frac{\rho_{\mathcal{R}}}{\rho_{b,A}} \approx 5.3, \quad \text{dark-to-baryonic abundance target,} \quad (25)$$

$$G\mu_{\text{visible}} \lesssim 10^{-7}, \quad \text{if horizon-scale visible strings are produced.} \quad (26)$$

Equation (26) is not a universal requirement for all compact relics. It applies if the transition produces a long-lived horizon-scale string network with visible gravitational/CMB signatures. Compact hidden-sector solitons face different constraints: self-interaction bounds, structure-formation limits, direct portal constraints, and stability requirements.

The relic idea is useful only if $\rho_{\mathcal{R}}$ is not chosen by hand. A predictive completion should produce a relation of the form

$$\frac{\rho_{\mathcal{R}}}{\rho_{b,A}} = \mathcal{F}[\Pi, T_\Pi, \mathcal{M}, \text{topology, couplings}], \quad (27)$$

and the observed value $\simeq 5.3$ should emerge as an output or a constrained fit.

6 Projection scale and the photon ratio

Let T_Π denote the temperature, curvature scale, or order-parameter condition at which the membrane projection becomes effective. The branch baryon residue is

$$\eta_A = \left. \frac{n_A^+ - n_A^-}{n_\gamma} \right|_{T \ll T_\Pi}. \quad (28)$$

The measured photon-to-baryon ratio then constrains the projection event. If projection is too early and too efficient, the model risks eliminating the annihilation history that produces the photon

bath. If projection is too late, it may simply project a residue whose size has already been fixed by conventional dynamics. In that case it explains only branch localization, not the value of η_A .

The useful late-projection scenario is:

- (1) The early state contains nearly balanced orientation sectors.
- (2) Thermal annihilation and plasma evolution create the photon bath.
- (3) At T_Π , the branch-polarization order parameter crystallizes or projects the remaining branch-relevant occupation measure into A and B .
- (4) A residual part of the projection stress-energy becomes $\mathcal{R}_\Pi$.

The model becomes predictive only if T_Π is fixed by an equation, for example

$$F[\Gamma_{\text{ann}}(T_\Pi), H(T_\Pi), m_\Pi, \lambda_\Pi, \xi_\Pi, \dots] = 0, \quad (29)$$

where $m_\Pi, \lambda_\Pi, \xi_\Pi$ are parameters of the order parameter or membrane sector. Otherwise T_Π is just another name for the observed residue.

7 A minimal effective-field-theory template

A symbolic EFT for the branch-separating order parameter can be written as

$$\mathcal{L}_\Pi = \frac{1}{2}(\partial_\mu \phi)^2 - V(\phi, T) + \mathcal{L}_{\text{portal}} + \mathcal{L}_{\text{top}}. \quad (30)$$

Here ϕ is a stand-in for the membrane polarization field; $V(\phi, T)$ produces a phase transition near T_Π ; $\mathcal{L}_{\text{portal}}$ controls any coupling to Standard Model or branch degrees of freedom; and $\mathcal{L}_{\text{top}}$ encodes topological terms responsible for relic stability.

A simple symmetry-breaking potential would be

$$V(\phi, T) = \frac{\lambda}{4} (\phi^2 - v^2(T))^2, \quad (31)$$

where $v(T) = 0$ above the transition and $v(T) \neq 0$ below it. The sign of ϕ can label two branch orientations,

$$\phi > 0 \rightarrow A, \quad \phi < 0 \rightarrow B. \quad (32)$$

Defects form where the order parameter cannot choose a single orientation globally. Depending on the topology of the vacuum manifold, these may be domain walls, strings, monopoles, textures, or compact solitons. Not all choices are viable. Stable domain walls are usually dangerous; visible cosmic strings are tightly constrained; compact hidden-sector solitons are less immediately excluded but must satisfy dark matter bounds.

The branch and relic yields may be summarized by efficiencies

$$Y_A = \frac{n_A^+ - n_A^-}{s}, \quad (33)$$

$$Y_B = -Y_A, \quad (34)$$

$$Y_{\mathcal{R}} = \frac{\rho_{\mathcal{R}}}{s}, \quad (35)$$

where s is entropy density. A real model must compute these from the transition, not fit them separately.

8 Why this is not ordinary antimatter-domain cosmology

If A and B are spatial domains in the same observable universe, the model faces annihilation-front constraints. Large adjacent matter and antimatter regions would produce gamma-ray and CMB signatures. This is not the intended interpretation.

The intended interpretation is branch separation:

$$\langle \psi_B | \mathcal{O}_{\text{SM}} | \psi_A \rangle \simeq 0 \quad (36)$$

for ordinary Standard Model observables. If this orthogonality is exact, the conjugate branch is not directly observable. If there is leakage,

$$\mathcal{A}_{A \rightarrow B} = \langle \psi_B | U | \psi_A \rangle \neq 0, \quad (37)$$

then the model becomes more testable and more constrained. A future version must specify whether the only common channel is gravity, a hidden portal, or no operational channel at all.

The relic sector changes this dilemma. Even if branch B is inaccessible, $\mathcal{R}_\Pi$ may remain gravitationally accessible to branch A . Thus the model need not rely on seeing the conjugate branch. Its empirical handle can be the relic abundance and relic phenomenology.

9 Relation to known frameworks

Spontaneous symmetry breaking and defects. The Π transition resembles a symmetry-breaking event. If the vacuum manifold has nontrivial topology, defects can form. This is familiar from cosmic strings, domain walls, monopoles, textures, and solitonic dark matter. The model must choose a viable defect class rather than invoke “crystallization” generically.

Intersecting branes and index structure. Intersecting-brane models provide a rigorous precedent for fermions localized at intersections and multiplicities controlled by topological data. If the membrane index I_A is meaningful, it should be translated into an intersection number, an index of a Dirac-type operator, or an anomaly-inflow relation.

Mirror and hidden sectors. The $A \oplus B$ structure resembles mirror-sector models. The novelty is not the existence of a second sector, but the possibility that A , B , and $\mathcal{R}_\Pi$ are co-produced by a single projection event.

Asymmetric dark matter. The ratio $\Omega_{\text{DM}}/\Omega_b \approx 5.3$ is suggestive because asymmetric-dark-matter models already try to relate the dark and baryonic abundances. The present model is different in mechanism but similar in ambition: the dark abundance should not be independent of the visible baryon residue.

Bell-type measurement dependence. A global membrane history may suggest measurement dependence in Bell-type experiments, but this is not part of Opus I. No joint distribution $\rho(\lambda, a, b)$ is derived here. Bell-related speculation should remain quarantined until the cosmological and particle-sector pieces are mathematically defined.

10 Predictions and falsifiability targets

The model is not predictive until it supplies at least one hard output. The most important targets are:

- (1) a derived membrane operator Π ;
- (2) a calculated projection scale T_Π ;
- (3) a branch residue $\eta_A \sim 6 \times 10^{-10}$;
- (4) a relic abundance $\rho_{\mathcal{R}}/\rho_b \approx 5.3$;
- (5) a relic mass, size, self-interaction, or power-spectrum signature;
- (6) a proof that $Q_{\text{SM}}(\mathcal{R}_\Pi) = 0$ and $Q_A = Q_B = 0$;
- (7) a constraint-compatible defect class, avoiding excluded visible string or domain-wall networks;
- (8) at least one relation among generation masses or mixings if the generation-projection analogy is retained.

The strongest possible future version would derive both η_A and $\Omega_{\text{DM}}/\Omega_b$ from the same topological or geometric data. That would turn the model from an interpretation into a physical proposal.

11 Conclusion

Opus I reformulates the polarized-membrane idea as a three-sector projection framework. The transition produces a visible matter-oriented branch A , a conjugate branch B , and a residual topological relic sector $\mathcal{R}_\Pi$. This reframing avoids two weak versions of the idea. It is not a local positron-exclusion rule, and it is not an ordinary matter–antimatter domain cosmology. Local QED and Standard Model gauge neutrality remain mandatory.

The new advantage is that the model gains a second quantitative target: the dark-to-baryonic matter ratio. If η_A and $\rho_{\mathcal{R}}/\rho_b$ are co-produced by the same membrane transition, their relation may encode real geometry or topology. If they are fitted independently, the model has only changed notation.

The model therefore lives or dies by the derivation of Π , T_Π , and the relic yield. This is the intended starting point for the series. Later notes should attempt to replace the phenomenological projectors with explicit membrane dynamics, index structure, and relic phenomenology.

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